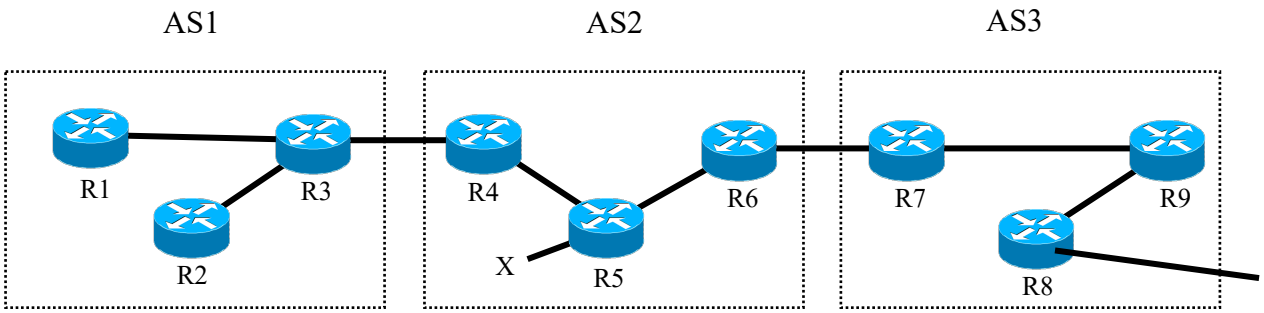
سؤال اول) شبکه زیر را که از سه autonomous system (AS) تشکیل شده است در نظر بگیرید. AS1 و AS2 برای مسیریابی داخلی از پروتکل OSPF و AS3 برای مسیریابی داخل از پروتکل RIP استفاده می‌کند. اطلاعات مسیریابی از طریق پروتکل‌های eBGP و iBGP بین ASها منتشر می‌شود. اطلاعات مربوط به پیشوند X (متصل به مسیریاب R5) از طریق کدام پروتکل‌ها به مسیریاب‌های R2، R7 و R8 می‌رسد؟



سوال 2) نکات مهم در متن زیر را بنویسید- ایا در متن اشتباه وجود دارد؟

For a congestion control algorithm to be useful in practice, it should be amenable to decentralized implementation. We now present one possible manner in which the primal algorithm could be implemented in a distributed fashion. We first note that each source

simply needs to know the sum of the link prices on its route to adjust its data rate as suggested by the algorithm. Suppose that every packet has a field (a certain number of bits) set aside in its header to collect the price of its route. When the source releases a packet into the network, the price field can be set to zero. Then, each link on the route can add its price to the price field so that, by the time the packet reaches its destination, the price field will contain the route price. This information can then be fed back to the source to implement the congestion control algorithm. A noteworthy feature of the congestion control algorithm is that the link prices depend only on the total arrival rate to the link, and not on the individual arrival rates of each source using the link. Thus, each link has only to keep track of the total arrival rate to compute the link price. If the algorithm required each link to keep track of individual source arrival rates, it would be infeasible to implement since the number of sources using high-capacity links could be prohibitively large. Thus, the primal algorithm is both amenable to a distributed implementation and has low overhead requirements.

Packet headers in the Internet are already crowded with a lot of other information, such as source/destination addresses to facilitate routing, so Internet practitioners do not like to add another field in the packet header to collect congestion information. In view of this, the overhead required to collect congestion information can be further reduced to accommodate practical realities. Consider the extreme case where there is only one bit available in the packet header to collect congestion information. Suppose that each packet is *marked* with probability $1 - e^{-p_l}$ when the packet passes through link l. Marking simply means that a bit in the packet header is flipped from a 0 to a 1 to indicate congestion. Then, along a route r, a packet is marked with probability

$$1 - e^{\sum_{l:l\in r} p_l}.$$

If the acknowledgement for each packet contains one bit of information to indicate whether a packet is marked or not, then, by computing the fraction of marked packets, the source can compute the route price $\sum_{l:l\in r} p_l$. The assumption here is that the x_r's change slowly so that each p_l remains roughly constant over many packets. Thus, one can estimate p_l approximately.

While marking, as mentioned above, has been widely studied in the literature, the predominant mechanism used for congestion feedback in the Internet today is through packet drops. Buffers used to store packets at a link have finite capacity, and therefore a packet that arrives at a link when its buffer is full is dropped immediately. If a packet is dropped, the destination of the packet will not receive it. So, if the destination then provides feedback to the source that a packet was not received, this provides an indication of congestion. Clearly, such a scheme does not require even a single bit in the packet header to collect congestion information. However, strictly speaking, this type of congestion feedback cannot be modeled using our framework since we assume that a source's data rate is seen by all links on the route, whereas, if packet dropping is allowed, some packets will not reach all links on a source's route. However, if we assume that the packet loss rate at each link is small, we can approximate the rate at which a link receives a source's packet by the rate at which the source is transmitting packets. Further, the end-to-end drop probability on a route can be approximated by the sum of the drop probabilities on the links along the route if the drop probability at each link is small. Thus, the optimization formulation approximates reality under these assumptions. To complete the connection to the optimization